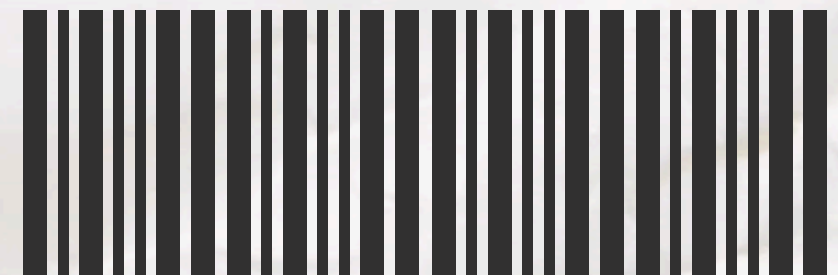
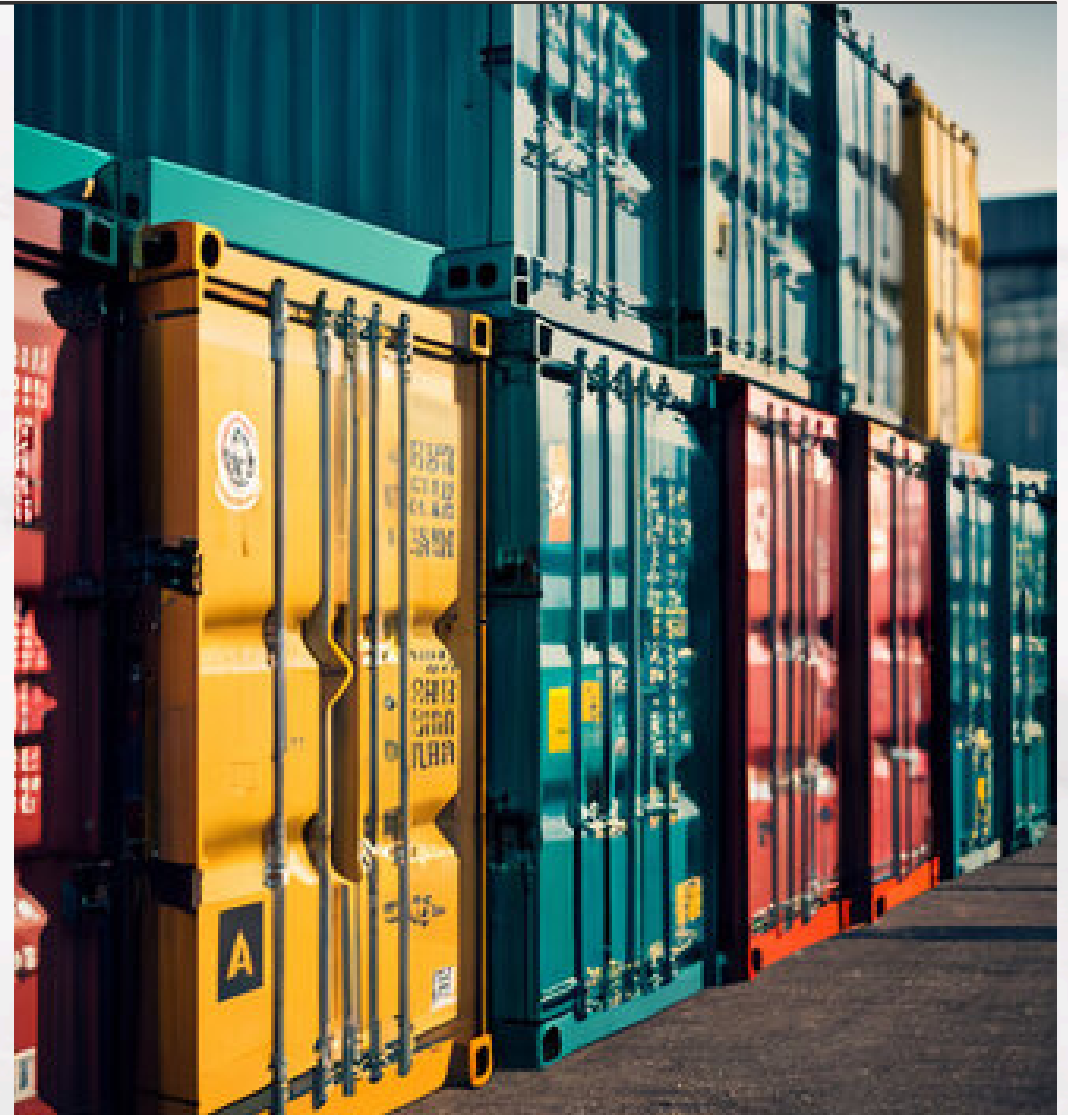


Byte Me

Gay Ming Kai, Chin Wei Hao, Christopher Lim Wai Ming



Byte Me's Executive Summary



Theme 1

An AI-driven solution to enhance port resilience and efficiency, reinforcing PSA's role as a reliable partner in global supply chain amidst increasing volatility.



Problem

Singapore's port is experiencing increased delays and longer vessel stays, causing supply chain disruptions and potential financial losses.



Relevancy to PSA

- Enhance port efficiency
- Enhance port resilience
- Ensure high system uptime
- In the rare event of system downtime
 - PSA Operation remain unaffected
 - Contingency plans ensure functionality without disruption

Our Team



Ming Kai



Wei Hao



Chris

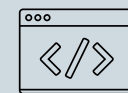


Solution



Front End:

An UI developed in Pygame to showcase the algorithm developed for route planning.



Back End:

ML models & algorithms to predict the route and the congestion of the ports



Data:

Open source-data from PSA website & naval topology

01. Problem statement

Singapore's port has experienced significant delays, with 90% of container vessels arriving off-schedule, compared to an average of about 77% in 2023

Vessel port stays at PSA have also increased by 22% compared to the same period last year

These disruptions have a cascading effect on our supply chains which can lead to increased costs, inefficiencies and potential revenue losses for shareholders



Problem vs. solution



Problem

Vessels that come off schedule
due to increase in demand and
container re-handling



Solution

An algorithm which suggest
container management using
other ports for effective
resource utilization

PSA Network

Connectivity

PSA has established itself all over the world

Reliability

PSA maintains delays through automation and AI-driven optimization.

Multimodal Services

PSA uses multiple modes of transportation and channels

Resilience

Node-To-Network enhances the flexibility of their global port network



PSA Process

01

Request

Submits a request for
shipping services

02

Pick-up

Arranges for container
pick-up from client's
facility

03

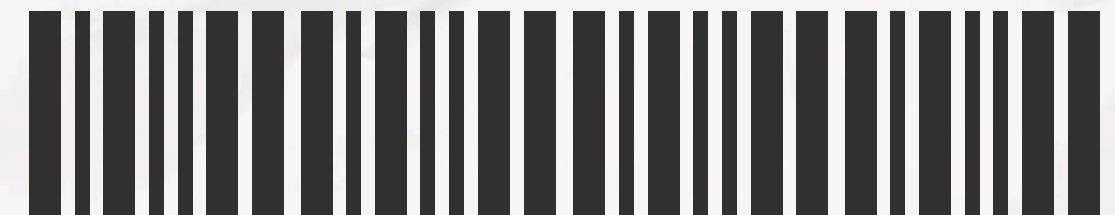
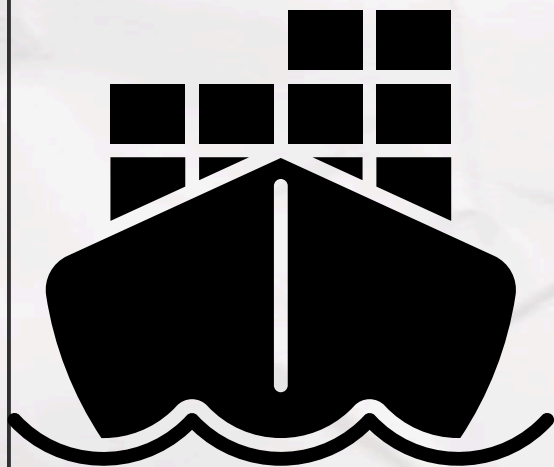
Process

Process shipment
details & calculate
optimal shipping routes

04

Suggest

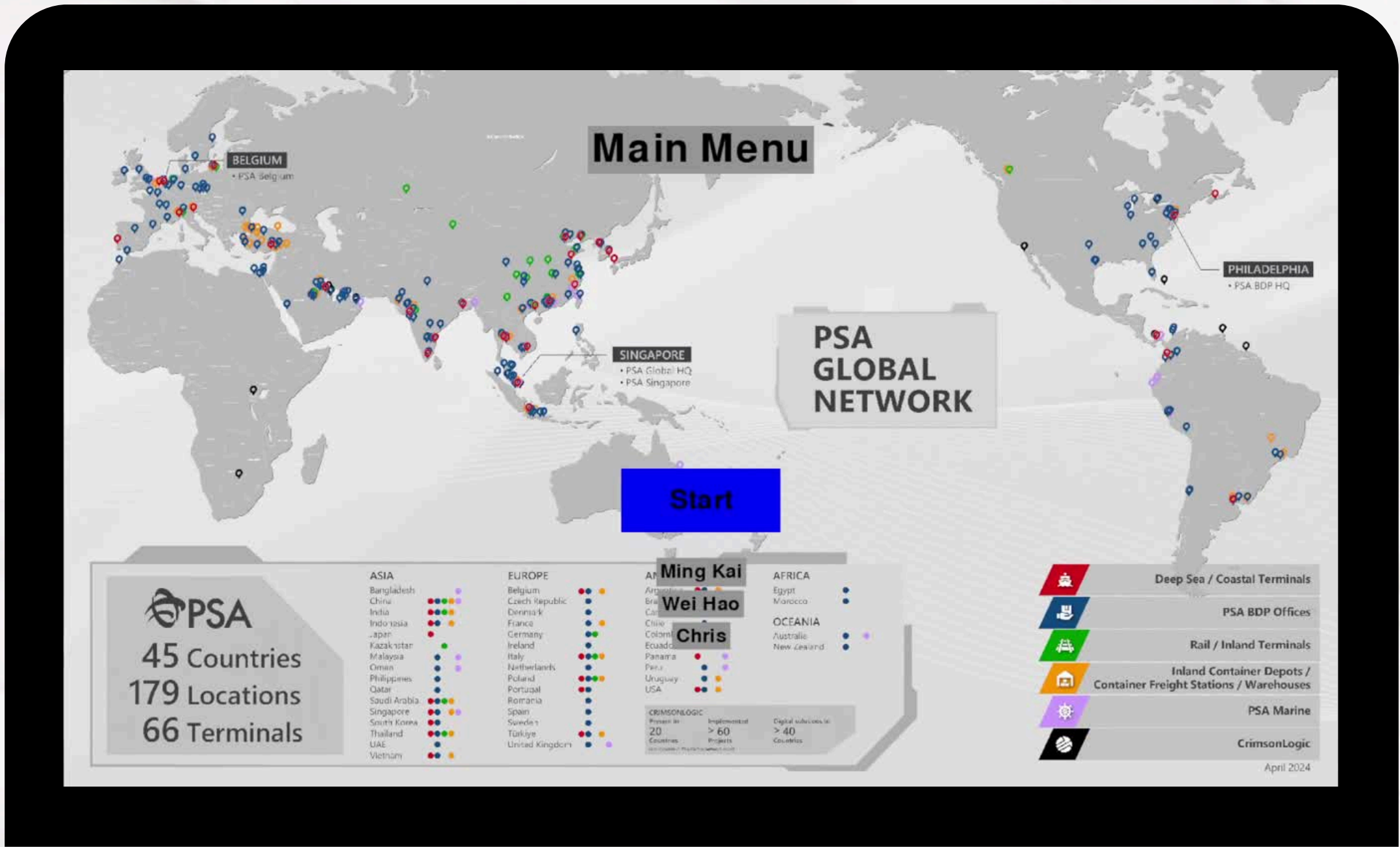
Provide the client with
optimal route and
delivery schedule



Our Solution (Prototype)

Components	Language	Description
Front-End	Python (Pygame)	Interactive UI showcasing the algorithm
Back-End	Python (Scikit Learn)	Logistic Regression predicting the weights
Data	Excel Spreadsheet	Obtained open source data of PSA Ports and Naval Topology

Working Example





Relevancy To PSA

Accurate: Real time data instead of Arbitrary data

Efficient: Allows for ports to be utilized to their fullest potential

Resilient: During crisis that can disrupt operations, the algorithm mitigates the damage

Future Timeline



Phase 1: Integration & Data Expansion

- Integrate with global ports
- More data collections and analysis



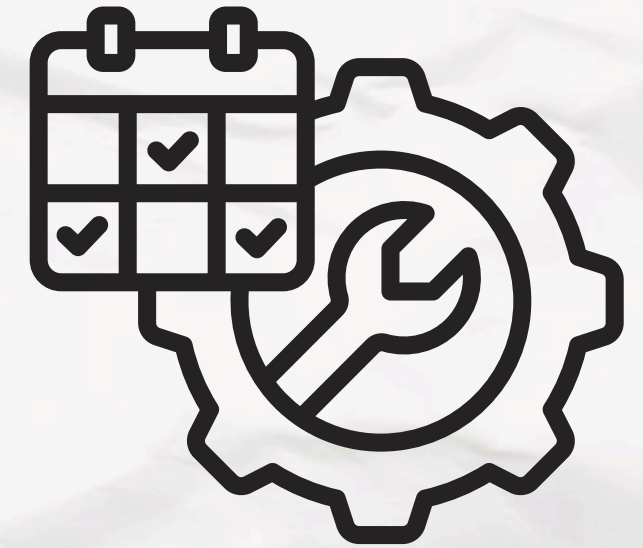
Phase 2: Advance Predictive Analytics

- Enhancing predictive accuracy
- Algorithm Optimization



Phase 3: PSA Leadership and Global Influence

- PSA as the leading innovator in AI-driven port management
- Setting industry standards



Phase 4: Predictive Maintenance and Ecosystem Expansion

- Expansion into predictive maintenance
- Global Port Network Ecosystem
- PSA as a hub

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